Science & Technology Group
Computer Networks
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Examen Declaratieve Talen

June 17th 2026 ViS

The exam lasts 3 hours.

1 Closed book

1.1 Data Link Layer

1.1.1 Question 1 (10 marks)

(a) Would listening or transmitting likely consume more energy on

i. a *star* topology?

ii. a *mesh* topology?

Be sure to explain your answer.

(b) How do MAC protocols ensure low energy consumption? Give concrete MAC protocol examples and explain how they achieve this.

1.1.2 Question 2 (5 marks)

Given is a satellite connection with bandwidth 3 Mbps, frame size 1500 B, and RTT 250 ms. Calculate the throughput for stop-and-wait.

Which protocol would you use to improve the throughput? Explain why.

1.2 Transport Layer

1.2.1 Question 1 (10 or 12 marks)

What specific features of RTP make it well suited for real-time live streaming applications, such as broadcasting a World Cup match to millions of viewers?

1.2.2 Question 2 (5 marks)

Draw TCP Tahoe congestion control graph. Annotate to explain the different phases.

1.2.3 Question 3 (5 marks)

Wat is ECN? Give one advantage and one disadvantage compared to Choke Packets and RED.

2 Open book

2.1 Application Layer

2.1.1 Question 1

How can we censor DNS traffic? Be specific about which layers and which devices would need to be involved.

2.1.2 Question 2

How would a copyright enforcement agency go about monitoring BitTorrent activity to detect illegal file sharing?

2.2 Network Layer

2.2.1 Question 1

Give 4 network layer concepts that are also used in Transport or Data Link layer. Why this redundancy? Is the overhead justified?

2.2.2 Question 2

What adaptations would be needed to run OSPF – which is normally used in LAN networks — on unreliable wireless nodes that do not move (e.g. IoT devices in a building)?